

3/4.7 PLANT SYSTEMS

3/4.7.1 TURBINE CYCLE

SAFETY VALVES

LIMITING CONDITION FOR OPERATION

3.7.1.1 All main steam line code safety valves (MSSVs) associated with each steam generator shall be OPERABLE with lift settings as specified in Table 3.7-4.

APPLICABILITY: MODES 1, 2 and 3.

ACTION:

- a. With one or two main steam line code safety valves inoperable in one or more steam generators, operation in Modes 1, 2 and 3 may proceed provided, that within 4 hours, either the inoperable valve is restored to OPERABLE status or reduce power to less than or equal to the applicable percent of RATED THERMAL POWER per Table 3.7-1; otherwise, be in at least HOT STANDBY within the next 6 hours and in HOT SHUTDOWN within the following 12 hours.
- b. With three main steam line code safety valves inoperable in one or more steam generators, operation in Modes 1, 2 and 3 may proceed provided, that within 4 hours, either the inoperable valves are restored to OPERABLE status or reduce power to less than or equal to the applicable percent of RATED THERMAL POWER per Table 3.7-1 and within 36 hours, reduce the Power Range Neutron Flux High trip setpoint to less than or equal to the RATED THERMAL POWER per Table 3.7-1; otherwise, be in at least HOT STANDBY within the next 6 hours and in HOT SHUTDOWN within the following 12 hours.

SURVEILLANCE REQUIREMENTS

4.7.1.1 Verify each required MSSV lift setpoint per Table 3.7-4. No additional Surveillance Requirements other than those required by the INSERVICE TESTING PROGRAM.

TABLE 3.7-1

MAXIMUM ALLOWABLE POWER WITH INOPERABLE
STEAM LINE SAFETY VALVES

<u>Maximum Number of Inoperable Safety Valves on Any Operating Steam Generator</u>	<u>Maximum Allowable Power* (Percent of RATED THERMAL POWER)</u>
1	87
2	58
3	39

* The values do not provide any allowance for calorimetric error.

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TABLE 3.7-4

STEAM LINE SAFETY VALVES PER LOOP

<u>VALVE NUMBER</u>					<u>LIFT SETTING ($\pm 3\%$)*</u>	<u>ORIFICE SIZE</u> (sq. inches)
	<u>Loop A</u>	<u>Loop B</u>	<u>Loop C</u>	<u>Loop D</u>		
a.	21MS11	22MS11	23MS11	24MS11	1125 psig	16.0
b.	21MS12	22MS12	23MS12	24MS12	1120 psig	16.0
c.	21MS13	22MS13	23MS13	24MS13	1110 psig	16.0
d.	21MS14	22MS14	23MS14	24MS14	1100 psig	16.0
e.	21MS15	22MS15	23MS15	24MS15	1070 psig	16.0

* Following testing the lift setting shall be reset to within $\pm 1\%$.

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AUXILIARY FEEDWATER SYSTEM

LIMITING CONDITION FOR OPERATION

3.7.1.2 At least three independent steam generator auxiliary feedwater pumps and associated manual activation switches in the control room and flow paths shall be OPERABLE with:

- a. Two feedwater pumps, each capable of being powered from separate vital busses, and
- b. One feedwater pump capable of being powered from an OPERABLE steam supply system.

APPLICABILITY: MODES 1, 2 and 3.

ACTION:

- a. With one auxiliary feedwater pump inoperable, restore the required auxiliary feedwater pumps to OPERABLE status within 72 hours or be in at least HOT STANDBY within the next 6 hours and in HOT SHUTDOWN within the following 6 hours.
- b. With two auxiliary feedwater pumps inoperable be in at least HOT STANDBY within 6 hours and in HOT SHUTDOWN within the following 6 hours.
- c. With three auxiliary feedwater pumps inoperable, immediately initiate corrective action to restore at least one auxiliary feedwater pump to OPERABLE status as soon as possible.
- d. LCO 3.0.4.b is not applicable.

SURVEILLANCE REQUIREMENTS

4.7.1.2 Each auxiliary feedwater pump shall be demonstrated OPERABLE:

- a. In accordance with the Surveillance Frequency Control Program by:
 1. Verifying that each non-automatic valve in the flow path that is not locked, sealed or otherwise secured in position, is in its correct position.
 2. Verify the manual maintenance valves in the flow path to each steam generator are locked open.

PLANT SYSTEMS

SURVEILLANCE REQUIREMENTS (Continued)

- b. In accordance with the Surveillance Frequency Control Program by:
 - 1. Verify that the developed head of each motor driven pump at the flow test point is greater than or equal to the required developed head.
 - 2. Verify that the developed head of the steam driven pump at the flow test point is greater than or equal to the required developed head when the steam generator pressure is > 680 psig. The provisions of Specification 4.0.4 are not applicable provided the surveillance is performed within 24 hours after secondary side pressure is greater than 680 psig.

- c. In accordance with the Surveillance Frequency Control Program by:
 - 1. Verifying that each auxiliary feedwater automatic valve that is not locked, sealed, or otherwise secured in position, actuates to the correct position on an actual or simulated actuation signal.
 - 2. Verifying that each auxiliary feedwater pump starts automatically on an actual or simulated actuation signal.

The provisions of Specification 4.0.4 are not applicable to the turbine driven auxiliary feedwater pump, provided the surveillance is performed within 24 hours after the secondary steam generator pressure is greater than 680 psig.

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AUXILIARY FEED STORAGE TANK

LIMITING CONDITION FOR OPERATION

3.7.1.3 The auxiliary feed storage tank (AFST) shall be OPERABLE with a contained volume of at least 200,000 gallons of water.

APPLICABILITY: MODES 1, 2 and 3.

ACTION:

With the auxiliary feed storage tank inoperable, within 4 hours either:

- a. Restore the AFST to OPERABLE status or be in HOT SHUTDOWN within the next 12 hours, or
- b. Demonstrate the OPERABILITY of a demineralized water or a fire protection/domestic water storage tank as a backup supply to the auxiliary feedwater pumps and restore the auxiliary feed storage tank to OPERABLE status within 7 days or be in HOT SHUTDOWN within the next 12 hours.

SURVEILLANCE REQUIREMENTS

4.7.1.3.1 The auxiliary feed storage tank shall be demonstrated OPERABLE in accordance with the Surveillance Frequency Control Program by verifying the water level is within its limits when the tank is the supply source for the auxiliary feedwater pumps.

4.7.1.3.2 A demineralized water storage tank shall be demonstrated OPERABLE in accordance with the Surveillance Frequency Control Program by verifying the tank contains greater than or equal to 200,000 gallons of water and by verifying proper alignment of valves for taking suction from this tank when it is the supply source for the auxiliary feedwater pumps.

4.7.1.3.3 A fire protection/domestic water storage tank shall be demonstrated OPERABLE in accordance with the Surveillance Frequency Control Program by verifying the tank contains greater than or equal to 200,000 gallons of water and by verifying proper alignment of valves for taking suction from this tank when it is the supply source for the auxiliary feedwater pumps.

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ACTIVITY

LIMITING CONDITION FOR OPERATION

3.7.1.4 The specific activity of the secondary coolant system shall be less than or equal to 0.10 $\mu\text{Ci/gram}$ DOSE EQUIVALENT I-131.

APPLICABILITY: MODES 1, 2, 3, and 4.

ACTION:

With the specific activity of the secondary coolant system greater than 0.10 $\mu\text{Ci/gram}$ DOSE EQUIVALENT I-131, be in at least HOT STANDBY within 6 hours and in COLD SHUTDOWN within the following 30 hours.

SURVEILLANCE REQUIREMENTS

4.7.1.4 The specific activity of the secondary coolant system shall be determined to be within the limit by performance of the sampling and analysis program of Table 4.7-2.

TABLE 4.7-2

SECONDARY COOLANT SYSTEM SPECIFIC ACTIVITY
SAMPLE AND ANALYSIS PROGRAM

<u>TYPE OF MEASUREMENT AND ANALYSIS</u>	<u>SAMPLE AND ANALYSIS FREQUENCY</u>
1. Gross Activity Determination	In accordance with the Surveillance Frequency Control Program
2. Isotopic Analysis for DOSE EQUIVALENT I-131 Concentration	a. In accordance with the Surveillance Frequency Control Program, whenever the gross activity determination indicates iodine concentrations greater than 10% of the allowable limit. b. In accordance with the Surveillance Frequency Control Program, whenever the gross activity determination indicates iodine concentrations below 10% of the allowable limit.

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MAIN STEAM LINE ISOLATION VALVES

LIMITING CONDITION FOR OPERATION

3.7.1.5 Each main steam line isolation valve shall be OPERABLE.

APPLICABILITY: MODE 1

MODES 2 and 3 except when all MSIVs are closed.

ACTION:

MODE 1 - With one main steam line isolation valve inoperable, POWER OPERATION may continue provided the inoperable valve is either restored to OPERABLE status or closed within 4 hours;

Otherwise, be in MODE 2 within the next 6 hours.

MODES 2 - With one or more main steam line isolation valve(s) inoperable, subsequent
and 3 operation in MODES 2 or 3 may proceed provided;

a. The isolation valve(s) is (are) maintained closed, and

b. The isolation valve(s) is (are) verified closed once per 7 days.

Otherwise, be in MODE 3, HOT STANDBY, within the next 6 hours, and
MODE 4, HOT SHUTDOWN, within the following 6 hours.

SURVEILLANCE REQUIREMENTS

4.7.1.5 Each main steam line isolation valve shall be demonstrated OPERABLE by verifying full closure within 5 seconds when tested pursuant to the INSERVICE TESTING PROGRAM. The provisions of Specification 4.0.4 are not applicable.

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3/4.7.2 STEAM GENERATOR PRESSURE/TEMPERATURE LIMITATION

LIMITING CONDITION FOR OPERATION

3.7.2 The temperatures of both the primary and secondary coolants in the steam generators shall be greater than 70°F when the pressure of either coolant in the steam generator is greater than 200 psig.

APPLICABILITY: At all times.

ACTION:

With the requirements of the above specification not satisfied:

- a. Reduce the steam generator pressure of the applicable side to less than or equal to 200 psig within 30 minutes, and
- b. Perform an engineering evaluation to determine the effect of the overpressurization on the structural integrity of the steam generator. Determine that the steam generator remains acceptable for continued operation prior to increasing its temperatures above 200°F.

SURVEILLANCE REQUIREMENTS

4.7.2 The pressure in each side of the steam generator shall be determined to be less than 200 psig in accordance with the Surveillance Frequency Control Program when the temperature of either the primary or secondary coolant is less than 70°F.

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3/4.7.3 COMPONENT COOLING WATER SYSTEM

LIMITING CONDITION FOR OPERATION

3.7.3 At least two independent component cooling water loops shall be OPERABLE.

APPLICABILITY: MODES 1, 2, 3 and 4.

ACTION:

With only one component cooling water loop OPERABLE, restore at least two loops to OPERABLE status within 72 hours or be in at least HOT STANDBY within the next 6 hours and in COLD SHUTDOWN within the following 30 hours.*

SURVEILLANCE REQUIREMENTS

4.7.3 At least two component cooling water loops shall be demonstrated OPERABLE:

- a. In accordance with the Surveillance Frequency Control Program by verifying that each valve (manual, power operated or automatic) servicing safety related equipment that is not locked, sealed, or otherwise secured in position, is in its correct position.

* The time for completion of repairs to number 22 component cooling water heat exchanger shall be extended from 0700 hours on November 23, 1982 to 1900 hours on November 23, 1982. If repairs are not completed by that time, the unit shall be placed in HOT STANDBY within the next 6 hours and in COLD SHUTDOWN within the following 30 hours.

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3/4.7.4 SERVICE WATER SYSTEM

LIMITING CONDITION FOR OPERATION

3.7.4 At least two independent service water loops shall be OPERABLE.

APPLICABILITY: MODES 1, 2, 3 and 4.

ACTION:

With only one service water loop OPERABLE, restore at least two loops to OPERABLE status within 72 hours or be in at least HOT STANDBY within the next 6 hours and in COLD SHUTDOWN within the following 30 hours.

SURVEILLANCE REQUIREMENTS

4.7.4 At least two service water loops shall be demonstrated OPERABLE:

- a. In accordance with the Surveillance Frequency Control Program by verifying that each valve (manual, power operated or automatic) servicing safety related equipment that is not locked, sealed, or otherwise secured in position, is in its correct position.
- b. In accordance with the Surveillance Frequency Control Program during shutdown, by verifying that each automatic valve servicing safety related equipment actuates to its correct position on Safeguards Initiation signal.

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3/4.7.5 FLOOD PROTECTION

LIMITING CONDITION FOR OPERATION

3.7.5 Flood protection shall be provided for all safety related systems, components and structures when the water level of the Delaware River exceeds 10.5' Mean Sea Level USGS datum, at the service water intake structure.

APPLICABILITY: At all times.

ACTION:

- a. With the water level at the service water intake structure above elevation 10.5' Mean Sea Level USGS datum, close all watertight doors within 2 hours.
- b. With the water level at the service water intake structure above elevation 11.5' Mean Sea Level USGS datum, be in at least HOT STANDBY within 6 hours and in COLD SHUTDOWN within the following 30 hours.

SURVEILLANCE REQUIREMENTS

4.7.5 The water level at the service water intake structure shall be determined to be within the limits by:

- a. Measurement in accordance with the Surveillance Frequency Control Program when the water level is below elevation 10.5' Mean Sea Level USGS datum, and
- b. Measurement in accordance with the Surveillance Frequency Control Program when the water level is equal to or above elevation 10.5' Mean Sea Level USGS datum.

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3/4.7.6 CONTROL ROOM EMERGENCY AIR CONDITIONING SYSTEM

LIMITING CONDITION FOR OPERATION

3.7.6 The common control room emergency air conditioning system (CREACS)* shall be OPERABLE with:

- a. Two independent air conditioning filtration trains (one from each unit) consisting of:
 1. Two fans and associated outlet dampers,
 2. One cooling coil,
 3. One charcoal adsorber and HEPA filter array,
 4. Return air isolation damper.
- b. All other automatic dampers required for operation in the pressurization or recirculation modes.
- c. The control room envelope intact.

NOTE: The control room envelope (CRE) boundary may be opened intermittently under administrative control.

APPLICABILITY: ALL MODES and during movement of irradiated fuel assemblies.

ACTION: MODES 1, 2, 3, and 4

- a. With one filtration train inoperable, align CREACS for single filtration train operation** within 4 hours, and restore the inoperable filtration train to OPERABLE status within 30 days or be in at least HOT STANDBY within the next 6 hours and in COLD SHUTDOWN within the following 30 hours.
- b. With CREACS aligned for single filtration train operation and with one of the two remaining fans or associated outlet damper inoperable, restore the inoperable fan or damper to OPERABLE status within 72 hours or be in at least HOT STANDBY within the next 6 hours and in COLD SHUTDOWN within the following 30 hours.
- c. With the Control Room Envelope boundary inoperable:
 1. Immediately, initiate action to implement mitigating actions, and
 2. Within 24 hours, verify mitigating actions ensure CRE occupant exposures to radiological, chemical, and smoke hazards will not exceed limits, and
 3. Within 90 days, restore the Control Room Envelope boundary to OPERABLE status, or be in at least HOT STANDBY within the next 6 hours and in COLD SHUTDOWN within the following 30 hours.

* The CREACS is a shared system with Salem Unit 1

** Alignment only permitted if the Unit with the operable CREACS train is also in Chilled Water System LCO 3.7.10a configuration. Alignment is not permitted if in the LCO 3.7.10c configuration.

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LIMITING CONDITION FOR OPERATION (Continued)

- d. With one or both series isolation damper(s) on a normal Control Area Air Conditioning System (CAACS) outside air intake or exhaust duct inoperable, close the affected duct within 4 hours by use of at least one isolation damper secured in the closed position or be in at least HOT STANDBY within the next 6 hours and in COLD SHUTDOWN within the following 30 hours. (Refer to ACTION 28 of Table 3.3-6.)
- e. With one or both isolation damper(s) on an outside emergency air conditioning air intake duct inoperable, close the affected duct within 4 hours by use of at least one isolation damper secured in the closed position and restore the damper(s) to OPERABLE status within 7 days or be in at least HOT STANDBY within the next 6 hours and in COLD SHUTDOWN within the following 30 hours.

MODES 5 and 6 or during movement of irradiated fuel assemblies

- a. With one filtration train inoperable, align CREACS for single filtration train operation** within 4 hours, or suspend movement of irradiated fuel assemblies.
- b. With CREACS aligned for single filtration train operation with one of the two remaining fans or associated outlet damper inoperable, restore the fan or damper to OPERABLE status within 72 hours, or suspend movement of irradiated fuel assemblies.
- c. With two filtration trains inoperable, immediately suspend movement of irradiated fuel assemblies.
- d. With the Control Room Envelope boundary inoperable, immediately suspend movement of irradiated fuel assemblies.
- e. With one or both series isolation damper(s) on a normal CAACS outside air intake or exhaust duct inoperable, immediately suspend movement of irradiated fuel assemblies until the affected duct is closed by use of at least one isolation damper secured in the closed position. (Refer to ACTION 28 of Table 3.3-6.)
- f. With one or both series isolation damper(s) on an outside emergency air conditioning air intake duct inoperable, immediately suspend movement of irradiated fuel assemblies until the affected duct is closed by use of at least one isolation damper secured in the closed position. To resume movement of irradiated fuel assemblies, at least one emergency air intake duct must be operable on each unit.

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SURVEILLANCE REQUIREMENTS

4.7.6.1 The control room emergency air conditioning system shall be demonstrated OPERABLE:

- a. In accordance with the Surveillance Frequency Control Program by initiating flow through the HEPA filter and charcoal adsorber train(s) and verifying that the train(s) operates with each fan operating for at least 15 minutes.
- b. In accordance with the Surveillance Frequency Control Program or prior to return to service (1) after any structural maintenance on the HEPA filter or charcoal adsorber housings, or (2) following painting, fire or chemical release in any ventilation zone communicating with the system, by:
 1. Verifying that the charcoal adsorbers remove $\geq 99\%$ of a halogenated hydrocarbon refrigerant test gas when they are tested in-place while operating the ventilation system at a flow rate of $8000 \text{ cfm} \pm 10\%$.
 2. Verifying that the HEPA filter banks remove $\geq 99\%$ of the DOP when they are tested in-place while operating the ventilation system at a flow rate of $8000 \text{ cfm} \pm 10\%$.
 3. Verifying within 31 days after removal from the CREACS unit, that a laboratory test of a sample of the charcoal adsorber, when obtained in accordance with Regulatory Position C.6.b of Regulatory Guide 1.52, Revision 2, March 1978, shows the methyl iodide penetration less than 2.5% when tested in accordance with ASTM D3803-1989 at a temperature of 30°C and a relative humidity of 95%.
- c. After every 720 hours of charcoal adsorber operation by verifying within 31 days after removal from the CREACS unit, that a laboratory analysis of a representative carbon sample, when obtained in accordance with Regulatory Position C.6.b of Regulatory Guide 1.52, Revision 2, March 1978, shows a methyl iodide penetration less than 2.5% when tested in accordance with ASTM D3803-1989 at a temperature of 30°C and a relative humidity of 95%.
- d. In accordance with the Surveillance Frequency Control Program by:
 1. Verifying that the pressure drop across the combined HEPA filter and charcoal adsorber bank is ≤ 3.5 inches water gauge while operating the ventilation system at a flow rate of $8000 \text{ cfm} \pm 10\%$.
 2. Verifying that on a safety injection test signal or control room intake high radiation test signal, the system automatically actuates in the pressurization mode by opening the outside air supply and diverting air flow through the HEPA filter and charcoal adsorber bank.
 3. Deleted.

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SURVEILLANCE REQUIREMENTS (Continued)

4. Verifying that on a manual actuation signal, the system will actuate to the required pressurization or recirculation operating mode.
 5. Verify each CREACS train has the capability to remove the assumed heat load.
 - e. After each complete or partial replacement of a HEPA filter bank by verifying that the HEPA filter banks remove $\geq 99\%$ of the DOP when they are tested in-place while operating the filter system at a flow rate of 8000 cfm $\pm 10\%$.
 - f. After each complete or partial replacement of a charcoal absorber bank by verifying that the charcoal absorbers remove $\geq 99\%$ of a halogenated hydrocarbon refrigerant test gas when they are tested in-place while operating the filter system at a flow rate of 8000 cfm $\pm 10\%$.
- 4.7.6.2 Perform required CRE unfiltered air inleakage testing in accordance with the Control Room Habitability Program (Refer to TS 6.17).

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3/4.7.7 AUXILIARY BUILDING VENTILATION SYSTEM

LIMITING CONDITION FOR OPERATION

3.7.7 At least two supply fans, and three exhaust fans shall be OPERABLE(*) to maintain the Auxiliary Building at slightly negative pressure.

-----NOTE-----

The intermittent opening of the Auxiliary Building pressure boundary causing a loss of negative pressure may be performed under administrative controls.

APPLICABILITY: At all times.

ACTION:

Modes 1 thru 4

- a) With one supply fan and/or one exhaust fan inoperable, restore the fan(s) to OPERABLE status within 14 days or be in HOT STANDBY within the next 6 hours and in COLD SHUTDOWN within the following 30 hours.
- b) With two supply and/or two exhaust fans inoperable restore at least one inoperable supply and two exhaust fans to operable status within 24 hours or be in HOT STANDBY within the next 6 hours and in COLD SHUTDOWN within the following 30 hours.
- c) With the Auxiliary Building pressure not maintained slightly negative, restore the building to slightly negative pressure within the next 4 hours or be in HOT STANDBY within the next 6 hours and in COLD SHUTDOWN within the following 30 hours.

During CORE ALTERATIONS

- d) With the Auxiliary Building pressure not maintained slightly negative, restore the Auxiliary Building to slightly negative pressure within the next 4 hours or suspend all operations involving CORE ALTERATIONS.

At all times

- e) With the Auxiliary Building pressure not maintained slightly negative, suspend all operations involving radioactive gaseous releases via the Auxiliary Building immediately.

(*) One of the supply fans may be considered OPERABLE with its auto start circuit administratively controlled (removed from service) to prevent more than one supply fan from operating at any time.

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SURVEILLANCE REQUIREMENTS

4.7.7 The above required Auxiliary Building Ventilation System shall be demonstrated OPERABLE:

- a) In accordance with the Surveillance Frequency Control Program by verifying negative pressure in the Auxiliary Building.
- b) In accordance with the Surveillance Frequency Control Program by starting each fan, from the control room, each fan operates for at least 15 minutes.
- c) In accordance with the Surveillance Frequency Control Program by verifying that the system starts following a Safety Injection Test Signal.

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SURVEILLANCE REQUIREMENTS

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3/4.7.8 SEALED SOURCE CONTAMINATION

LIMITING CONDITION FOR OPERATION

3.7.8 Each sealed source containing radioactive material either in excess of 100 microcuries of beta and/or gamma emitting material or 5 microcuries of alpha emitting material shall be free of greater than or equal to 0.005 microcuries of removable contamination.

APPLICABILITY: At all times.

ACTION:

- a. With a sealed source having removable contamination in excess of the above limits, immediately withdraw the sealed source from use and:
 - 1. Either decontaminate and repair the sealed source, or
 - 2. Dispose of the sealed source in accordance with Commission Regulations.
- b. The provisions of Specification 3.0.3 are not applicable.

SURVEILLANCE REQUIREMENTS

4.7.8.1 Test Requirements - Each sealed source shall be tested for leakage and/or contamination by:

- a. The licensee, or
- b. Other persons specifically authorized by the Commission or an Agreement State.

The test method shall have a detection sensitivity of at least 0.005 microcuries per test sample.

4.7.8.2 Test Frequencies - Each category of sealed sources (excluding startup sources and fission detectors previously subjected to core flux) shall be tested at the frequency described below.

- a. Sources in use - In accordance with the Surveillance Frequency Control Program for all sealed sources containing radioactive materials:

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SURVEILLANCE REQUIREMENTS (Continued)

1. With a half-life greater than 30 days (excluding Hydrogen 3),
and
 2. In any form other than gas.
- b. Stored sources not in use - Each sealed source and fission detector shall be tested prior to use or transfer to another licensee unless tested within the previous six months. Sealed sources and fission detectors transferred without a certificate indicating the last test date shall be tested prior to being placed into use.
- c. Startup sources and fission detectors - Each sealed startup source and fission detector shall be tested within 31 days prior to being subjected to core flux and following repair or maintenance to the source or detector.

4.7.8.3 Reports - A report shall be prepared and submitted to the Commission on an annual basis if sealed source or fission detector leakage tests reveal the presence of greater than or equal to 0.005 microcuries of removable contamination.

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3/4.7.9 SNUBBERS

LIMITING CONDITION FOR OPERATION

3.7.9 All snubbers shall be OPERABLE.

APPLICABILITY: MODES 1, 2, 3 and 4. (MODES 5 and 6 for snubbers located on systems required OPERABLE in those MODES).

ACTION:

With one or more snubbers inoperable, within 72 hours, replace or restore the inoperable snubber(s) to OPERABLE status and perform an engineering evaluation per Specification 4.7.9 on the supported component or declare the supported system inoperable and follow appropriate ACTION statement for that system.

SURVEILLANCE REQUIREMENTS

4.7.9 Each snubber shall be demonstrated OPERABLE by performance of the Snubber Testing Program for examination, testing and service life monitoring of snubbers.

Pages 3/4 7-25 through 3/4 7-27 have been deleted.

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3/4.7.10 CHILLED WATER SYSTEM - AUXILIARY BUILDING SUBSYSTEM

LIMITING CONDITION FOR OPERATION

3.7.10 The chilled water system loop which services the safety-related loads in the Auxiliary Building shall be OPERABLE with one of the following configurations:

	a	b	c
Configuration	1. Three OPERABLE chillers and, 2. Two OPERABLE chilled water pumps	1. Two OPERABLE chillers and, 2. Two OPERABLE chilled water pumps	1. Three OPERABLE chillers and, 2. Two OPERABLE chilled water pumps from either Unit 1 or Unit 2 (Units Cross-tied) ⁽²⁾
APPLICABILITY	1. ALL MODES and during movement of irradiated fuel assemblies	1. From November 1 through April 30 in ALL MODES and during movement of irradiated fuel assemblies [#] 2. The Unit 2 Emergency Control Air Compressor (ECAC) is isolated from the chilled water system 3. Chilled water flow to the third chiller that is not in service is isolated ⁽¹⁾ 4. Control Room Emergency Air Conditioning System (CREACS) alignment a. BOTH CREACS trains OPERABLE, no additional chilled water heat load removal required, OR b. Single CREACS train OPERABLE (TS 3.7.6 ACTION a.) the following restrictions apply: i. Alignment only permitted to Unit 1 ii. Unit 1 must be in the LCO 3.7.10a configuration iii. Non-essential heat loads are isolated from the chilled water system on BOTH Units	1. From November 1 through April 30 in ALL MODES and during movement of irradiated fuel assemblies ^{##} 2. The Unit 1 and Unit 2 ECACs are isolated from the chilled water system 3. Non-Essential heat loads are isolated from the chilled water system on BOTH Units 4. BOTH CREACS trains are operable per TS 3.7.6 (single filtration train alignment is not permitted) 5. Unit chilled water cross-tie valves are OPEN 6. Administrative controls are in place for the Unit providing the required components to notify the other Unit if a chiller or pump becomes inoperable

The LCO 3.7.10b configuration may only be used for periods of 60 contiguous days. The 60-contiguous days does not apply for LCO 3.7.10b entry to support the replacement of all 6 original chillers (Units 1 and 2).

The LCO 3.7.10c configuration may only be used for periods of 45 contiguous days.

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LIMITING CONDITION FOR OPERATION (Continued)

ACTION⁽³⁾: MODES 1, 2, 3, and 4

- a. With one of the required chillers inoperable:
 - 1. Remove⁽⁴⁾ the appropriate non-essential heat loads from the chilled water system within 4 hours and;
 - 2. Restore the chiller to OPERABLE status within 14 days or;
 - 3. Be in at least HOT STANDBY within the next 6 hours and in COLD SHUTDOWN within the following 30 hours.
- b. With two of the required chillers inoperable⁽⁵⁾⁽⁶⁾:
 - 1. Remove the appropriate non-essential heat loads from the chilled water system within 4 hours and;
 - 2. Align the control room emergency air conditioning system (CREACs) for single filtration operation using the Salem Unit 1 train within 4 hours and;
 - 3. Restore at least one chiller to OPERABLE status within 72 hours or;
 - 4. Be in at least HOT STANDBY within the next 6 hours and in COLD SHUTDOWN within the following 30 hours.
- c. With one chilled water pump inoperable, restore the chilled water pump to OPERABLE status within 7 days or be in at least HOT STANDBY within the next 6 hours and in COLD SHUTDOWN within the following 30 hours

ACTION⁽³⁾: MODES 5 and 6 or during movement of irradiated fuel assemblies.*

- a. With one of the required chillers inoperable:
 - 1. Remove⁽⁴⁾ the appropriate non-essential heat loads from the Chilled Water System within 4 hours and;
 - 2. Restore the chiller to OPERABLE status within 14 days or;
 - 3. Suspend CORE ALTERATIONS and movement of irradiated fuel assemblies.

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LIMITING CONDITION FOR OPERATION (Continued)

- b. With two of the required chillers inoperable⁽⁵⁾⁽⁶⁾:
 - 1. Remove the appropriate non-essential heat loads from the chilled water system within 4 hours and;
 - 2. Align the control room emergency air conditioning system (CREACs) for single filtration operation using the Salem Unit 1 train within 4 hours and;
 - 3. Restore at least one chiller to OPERABLE status within 72 hours or;
 - 4. Suspend CORE ALTERATIONS and movement of irradiated fuel assemblies.
- c. With one chilled water pump inoperable, restore the chilled water pump to OPERABLE status within 7 days or suspend CORE ALTERATIONS and movement of irradiated fuel assemblies.

SURVEILLANCE REQUIREMENTS

4.7.10 The chilled water loop which services the safety-related loads in the Auxiliary Building shall be demonstrated OPERABLE:

- a. In accordance with the Surveillance Frequency Control Program by verifying that each manual valve in the chilled water system flow path servicing safety related components that is not locked, sealed, or otherwise secured in position, is in its correct position.
- b. In accordance with the Surveillance Frequency Control Program, by verifying that each automatic valve actuates to its correct position on a Safeguards Initiation signal.
- c. In accordance with the Surveillance Frequency Control Program by verifying that each chillers starts and runs.
- d. When in the LCO 3.7.10b configuration verify once per 24 hours:
 - (i) The Unit 2 ECAC is isolated from the chilled water system,
 - (ii) Chilled water flow is isolated to the third chiller that is not in service and,
 - (iii) If CREACS is in single filtration alignment verify non-essential heat loads are isolated from the chilled water system on BOTH Units.
- e. When in the LCO 3.7.10c configuration verify once per 24 hours:
 - (i) The Unit 1 and Unit 2 ECACs are isolated from the chilled water system,
 - (ii) Non-essential heat loads are isolated from the chilled water system and,
 - (iii) Cross-tie valves are verified OPEN.

PLANT SYSTEMS

Surveillance Requirements (Continued)

NOTES

- (1) When transitioning from the LCO 3.7.10b to the LCO 3.7.10a configuration, the chiller may be un-isolated (restored to service) under administrative controls
 - (2) The LCO 3.7.10c (Cross-Tied) configuration is common to both Units; either Unit 1 chilled water components are required operable, OR Unit 2. A combination of both Units chilled water components is not permitted. When transitioning from the LCO 3.7.10c configuration to either the LCO 3.7.10a or LCO 3.7.10b configurations, chilled water components may be restored to service under administrative controls
 - (3) When in the LCO 3.7.10c configuration ACTIONS are applicable for both Units
 - (4) When in the LCO 3.7.10c configuration this ACTION has already been implemented
 - (5) When in the LCO 3.7.10b configuration, implement Action b.2 AND Action b.4 OR transition to the LCO 3.7.10c configuration
 - (6) When in LCO 3.7.10c configuration, proceed directly to Action b.4
- * During Modes 5 and 6 and during movement of irradiated fuel assemblies, chilled water components are not considered to be inoperable solely on the basis that the backup emergency power source, diesel generator, is inoperable. This is not applicable to the LCO 3.7.10c configuration.

PLANT SYSTEMS

3/4.7.11 FUEL STORAGE POOL BORON CONCENTRATION

LIMITING CONDITION FOR OPERATION

3.7.11 The fuel storage pool boron concentration shall be ≥ 800 ppm

APPLICABILITY: When fuel assemblies are stored in the fuel storage pool and a fuel storage pool verification has not been performed since the last movement of fuel assemblies in the fuel storage pool.

ACTION:

With fuel storage pool boron concentration not within limit:

- a. Immediately suspend movement of fuel assemblies in the fuel storage pool and
- b. Initiate action to:
 1. immediately restore fuel storage pool boron concentration to within limit or
 2. immediately perform a fuel storage pool verification.
- c. The provisions of Specification 3.0.3 is not applicable.

SURVEILLANCE REQUIREMENTS

4.7.11 Verify the fuel storage pool boron concentration is within limit in accordance with the Surveillance Frequency Control Program.

PLANT SYSTEMS

3/4.7.12 FUEL ASSEMBLY STORAGE IN THE SPENT FUEL POOL

LIMITING CONDITION FOR OPERATION

3.7.12 The combination of initial enrichment, burnup, and Integral Fuel Burnable Absorber (IFBA) of each fuel assembly stored in Region 1 or Region 2, shall be within the acceptable limits described in the surveillance requirements below.

APPLICABILITY: When any fuel assembly is stored in Region 1 or Region 2 of the spent fuel storage pool.

ACTION:

If the requirements of the LCO are not met:

- a. Immediately verify the fuel storage boron concentration meets the requirements of TS 3.7.11 and
- b. Immediately initiate action to move the non-complying fuel assembly to a location that complies with the surveillance requirements.
- c. The provisions of Specification 3.0.3 are not applicable.

SURVEILLANCE REQUIREMENTS

4.7.12.1 Prior to storing fuel assemblies in Region 1, verify by administrative means that the fuel assemblies meet one of the following storage constraints:

- a. Unirradiated fuel assemblies with a maximum enrichment of 4.25 wt% U-235 have unrestricted storage.
- b. Unirradiated fuel assemblies with enrichments greater than 4.25 wt% U-235 and less than or equal to 5.0 wt% U-235, that do not contain IFBA pins, may only be stored in the peripheral cells facing the concrete wall.
- c. Unirradiated fuel assemblies with enrichments (E) greater than 4.25 wt% U-235 and less than or equal to 5.0 wt% U-235, which contain a minimum number of IFBA pins have unrestricted storage. This minimum number of IFBA pins shall have an equivalent reactivity hold-down which is greater than or equal to the reactivity hold-down associated with N IFBA pins, at a nominal 2.35 mg B-10/linear inch loading (1.5x), determined by the equation below:

$$N = 42.67 (E - 4.25)$$

PLANT SYSTEMS

SURVEILLANCE REQUIREMENTS (continued)

- d. Irradiated fuel assemblies with enrichments (E) greater than 4.25 wt% U-235 and less than or equal to 5.0 wt%, that have attained the minimum burnup (BU) as determined by the equation below, have unrestricted storage.

$$BU \text{ (MWD/kg U)} = -26.212 + 6.1677E$$

4.7.12.2 Prior to storing fuel assemblies in Region 2, verify by administrative means that the fuel assemblies meet one of the following storage constraints:

- a. Unirradiated fuel assemblies with a maximum enrichment of 5.0 wt% U-235 may be stored in a checkerboard pattern with intermediate cells containing only water or non-fissile bearing material.
- b. Unirradiated fuel assemblies with a maximum enrichment (E) of 5.0 wt% U-235 may be stored in the central cell of any 3x3 array of cells provided the surrounding eight cells are empty or contain fuel assemblies that have attained the minimum burnup (BU) as determined by the equation below.

$$BU \text{ (MWD/kg U)} = -15.48 + 17.80E - 0.7038E^2$$

In this configuration, none of the nine cells in any 3x3 array shall be common to cells in any other similar 3x3 array. Along the rack periphery, the concrete wall is equivalent to 3 outer cells in a 3x3 array.

- c. Irradiated fuel assemblies with a maximum enrichment (E) of 5.0 wt% U-235 that have attained the minimum burnup (BU) as determined by the equation below, have unrestricted storage.

$$BU \text{ (MWD/kg U)} = -32.06 + 25.21E - 3.723E^2 + 0.3535E^3$$

- d. Irradiated fuel assemblies with a maximum enrichment (E) of 5.0 wt% U-235 that have attained the minimum burnup (BU) as determined by the equation below, may be stored in a peripheral cell facing the concrete wall.

$$BU \text{ (MWD/kg U)} = -25.56 + 15.14E - 0.602E^2$$

PLANT SYSTEMS

3/4.7.13 MAIN FEEDWATER ISOLATION VALVES (FIVS), MAIN FEEDWATER REGULATING VALVES (FRVS), FRV BYPASS VALVES (FRVBVS), AND STEAM GENERATOR FEEDWATER PUMP (SGFP) TURBINE STEAM STOP VALVES

LIMITING CONDITION FOR OPERATION

3.7.13 Four Main FIVs, four Main FRVs, four Main FRVBV, and four SGFP turbine steam stop valves shall be OPERABLE.

APPLICABILITY:

For the FIV in each main feedwater line:

MODES 1, 2, and 3 except when:

- a. The FIV is closed and deactivated; or
- b. The associated FRV and FRVBV are closed and deactivated; or
- c. The associated main feedwater line is isolated by a closed manual valve

For the FRV in each main feedwater line:

MODES 1, 2, and 3 except when:

- a. The FRV is closed and deactivated; or
- b. The associated FIV is closed and deactivated; or
- c. The associated main feedwater line is isolated by a closed manual valve

For the FRVBV in each main feedwater line:

MODES 1, 2, and 3 except when:

- a. The FRVBV is closed and deactivated; or
- b. The associated FIV is closed and deactivated; or
- c. The associated main feedwater line is isolated by a closed manual valve

For each SGFP Turbine Steam Stop Valve:

MODES 1, 2, and 3 except when:

- a. The SGFP Turbine Steam Stop Valve is closed and deactivated; or
- b. The associated steam supply to the SGFP turbine is isolated by a closed manual valve; or
- c. The SGFP feedwater flow path is isolated

PLANT SYSTEMS

3/4.7.13 MAIN FEEDWATER ISOLATION VALVES (FIVS), MAIN FEEDWATER REGULATING VALVES (FRVS), FRV BYPASS VALVES (FRVBVS), AND STEAM GENERATOR FEEDWATER PUMP (SGFP) TURBINE STEAM STOP VALVES

LIMITING CONDITION FOR OPERATION (continued)

ACTION:

NOTE

Separate Condition Entry is allowed for each valve

- a. With one or more FIVs inoperable, restore the inoperable FIV(s) to OPERABLE status or close or isolate the inoperable FIV(s) within 72 hours; verify the inoperable FIV(s) is closed or isolated once per 7 days.
- b. With one or more FRVs inoperable, restore the inoperable FRV(s) to OPERABLE status or close or isolate the inoperable FRV(s) within 72 hours; verify the inoperable FRV(s) is closed or isolated once per 7 days.
- c. With one or more FRVBV(s) inoperable, restore the inoperable FRVBV(s) to OPERABLE status or close or isolate the inoperable FRVBV(s) within 72 hours; verify the inoperable FRVBV(s) is closed or isolated once per 7 days.
- d. With one or more SGFP turbine steam stop valves inoperable, restore the inoperable SGFP turbine stop valve(s) to OPERABLE status or isolate the associated steam supply to the SGFP turbine or isolate the SGFP flow path within 72 hours; verify that the inoperable SGFP steam stop valve is isolated or the SGFP flow path is isolated once per 7 days.
- e. With two (2) valves in the same feedwater flowpath inoperable resulting in a loss of feedwater isolation capability for a flow path, restore at least one valve to OPERABLE status or isolate the affected flow path within 8 hours.
- f. With the required ACTION requirements above not met, be in HOT STANDBY within 6 hours and in HOT SHUTDOWN within the following 6 hours.

SURVEILLANCE REQUIREMENTS

- 4.7.13.1 Each FIV, FRV, FRVBV and SGFP turbine steam stop valve shall be demonstrated OPERABLE by determining the isolation time of each valve to be within limits when tested pursuant to the INSERVICE TESTING PROGRAM.
- 4.7.13.2 In accordance with the Surveillance Frequency Control Program, verify each FIV, FRV, FRVBV and SGFP turbine steam stop valve actuates to the isolation position on an actual or simulated actuation signal.